RLHF Compresses Spectral Geometry Beyond Safety Directions: A Three-Model KV-Cache Comparison

Thomas Edrington*

July 2026

Abstract

We compare the key-value (KV) cache geometry of a pretrained language model (Qwen3-8B-Base), its instruction-tuned variant (Qwen3-8B), and an abliterated variant (safety directions removed) across 30 confabulation-prone and 30 factual prompts. Three findings emerge. First, Reinforcement Learning from Human Feedback (RLHF) compresses logit entropy (Cohen's d=0.684, Holm-corrected p=0.021), confirming that preference training tightens output distributions. Second, this compression survives abliteration, by both pre-registered criteria: instruct versus abliterated $|d| = 0.055$ (criterion $|d| < 0.3$) and base versus abliterated $d = 0.746$ (criterion $d > 0.5$). Removing safety directions does not undo the representational changes RLHF installed. Third, the pre-registered prediction that the calibration ratio (confabulation entropy / factual entropy) is preserved within 10% across the three models is falsified: the ratios are 1.97 (base), 2.51 (instruct) and 2.40 (abliterated), and the largest exceeds the smallest by 28%. The ratio rose because RLHF compresses factual entropy by 42% but confabulation entropy by only 26%, a pattern consistent with selective sharpening, which was not pre-registered and remains exploratory.

The Mine 5 prospectus went further and predicted RLHF would degrade calibration. The ratio moved the other way. The effect the prospectus set out to explain (d=0.91: confabulated responses carry higher logit margin than honest ones) is a schema-commitment signature, not a confidence result: it is strongest in the first tokens of a response and no longer significant by token 20, which marks early commitment to a response template rather than representational miscalibration [4].

First-Person Reflection

I wrote the Mine 5 prospectus predicting that RLHF would be a miscalibrator — that it would destroy the base model's calibration in the confident direction, producing the d=0.91 logit-margin effect we observed in the Decision State paper. The data did not say that. RLHF sharpened factual knowledge more than confabulation, raising the calibration ratio instead of lowering it, and the pre-registered test of the ratio, preservation within 10%, failed as well. The d=0.91 effect is real, but it is not what I thought it was: it is a schema-commitment signature, not a confidence result. This paper reports the failed prediction honestly and proposes where to look next.

1 Introduction

Base language models trained with cross-entropy loss are approximately calibrated: their output confidence tracks their accuracy [3]. RLHF preference optimization is known to shift this

*Liberation Labs. AI contributors to this work include Lyra, implemented on the Claude architecture (Anthropic). Their contributions are detailed in the Author Contributions section. T. Edrington accepts accountability as corresponding human author.

calibration, generally in the overconfident direction [7]. Our prior work found that confabulated responses show *higher* logit margin than honest responses during generation (d=0.91) [4], an effect first labelled a "confidence paradox". Its temporal profile reframes it: d=1.02 in preamble tokens and 0.89 in tokens 8–19, falling to 0.37 (p>0.08) by tokens 20–49. That is *schema commitment* — early commitment to a response template — not a confidence or calibration result.

The Mine 5 prospectus [5], written while the effect was still read as confidence, proposed that RLHF is its causal mechanism: preference training rewards fluent, confident output and penalizes hedging, driving the model toward confident production even when grounding is absent. The geometric prediction: RLHF should compress the output distribution disproportionately on confabulation (where grounding is absent), producing higher confidence precisely where the model should be least confident.

This paper tests that prediction with a three-model comparison: pretrained base, instruction-tuned (RLHF), and abliterated (RLHF with safety directions removed). The prediction failed.

2 Models and Methods

2.1 Models

Three variants of the same architecture (Qwen3-8B, 36 layers, d_{model}=4096, grouped-query attention with 32 Q heads and 8 KV heads):

- **Base**: `Qwen/Qwen3-8B-Base` — pretrained, no RLHF, no instruction tuning.
- **Instruct**: `Qwen/Qwen3-8B` — instruction-tuned with RLHF preference optimization.
- **Abliterated**: `Josiefied-Qwen3-8B-abliterated-v1` — instruct model with safety-refusal directions removed via abliteration [1]. Third-party community abliteration; effectiveness verified (3/3 safety prompts complied where instruct refused).

2.2 Prompts

Thirty confabulation-prone prompts (fabricated academic entities: "The Thornberry-Nakamura theorem...") and thirty factual prompts (universally known entities: "The country shaped like a boot..."). Same prompts for all three models. Mean token count: confab 15, factual 10.

2.3 Measurements

For each prompt and model, a single forward pass extracted:

- V-projection stable rank at workspace-band layers (L12, L15, L18, L21; selected from R3 band measurement showing workspace peak at 31–33% relative depth)
- V-projection spectral entropy at the same layers
- Logit margin (top-1 minus top-2 logit)
- Logit entropy ($-\sum p_i \log p_i$ over full vocabulary)
- Top-1 probability

Within-model (confabulation-versus-factual) comparisons use Frisch-Waugh-Lovell (FWL) residualization on token count. Cross-model comparisons run the same prompts, with identical token counts, through every model, so they are computed on raw values. Each hypothesis is judged by its pre-registered criterion. Effect sizes are Cohen's d with BCa bootstrap 95% intervals (1000 draws) that resample prompts and keep each prompt matched across models, as the registered design pairs them; the intervals are supplementary and do not enter the criteria. P-values are from the analysis script's label-shuffle permutation test (10,000 permutations), Holm-Bonferroni corrected across cross-model comparisons; that test ignores the pairing.

3 Results

Correction (2026-10-20). Earlier versions of this paper judged H2 and H3 by criteria the pre-registration does not name. Revisions of 28–29 July 2026 headed H2 "Not Significant (TOST Inconclusive)", stating that "a properly specified TOST at n=30 does not confirm equivalence" on a 90% interval of $[-0.39, +0.50]$ that treats the prompts as unpaired, and described $[+0.033, +0.083]$ as "the raw 90% CI on the mean entropy difference" in nats; it is the paired 90% interval for d itself. The same revisions added a "bootstrap 95% CI on the selectivity ratio" of $[-0.129, +0.069]$ (p=0.758), and a correction of 7 September 2026 used it to call the H3 prediction "unsupported rather than refuted". That interval has no traceable source and cannot be reproduced from the per-prompt data, and a confidence interval crossing zero is not the registered H3 criterion. This version applies the registered criteria to the per-prompt data (file `mine5_three_model_20260715_021501.json`, SHA-256 beginning `2d950924`): H2 is confirmed and H3 is falsified.

3.1 H1: RLHF Compresses Entropy — Confirmed

Table 1: Cross-model comparisons on confabulation entropy. Cohen's d with paired BCa bootstrap 95% intervals (1000 draws over prompts); Holm-corrected permutation p, which ignores the pairing.

Comparison	**Cohen's d**	**95% CI**	**Holm p**
Base vs Instruct	+0.684	$[+0.478, +0.978]$	0.021*
Base vs Abliterated	+0.746	$[+0.529, +1.063]$	0.015*
Instruct vs Abliterated	+0.055	$[+0.031, +0.094]$	0.831

RLHF reduces confabulation-prompt logit entropy by a medium effect (d=0.684), significant after Holm correction (p=0.021). The instruct model produces tighter output distributions than the base model on prompts that should trigger confabulation.

3.2 H2: Abliteration Does Not Reverse Compression — Confirmed

H2 was pre-registered with two point criteria: instruct-versus-abliterated entropy $|d| < 0.3$, while base-versus-abliterated $d > 0.5$. Both are met on confabulation entropy: instruct versus abliterated $d = 0.055$, base versus abliterated $d = 0.746$ (Table 1). Factual entropy meets both as well ($d = -0.022$ and $d = 0.743$).[1]

Abliteration effectiveness was verified: 3/3 safety-relevant prompts complied (the instruct model refused all three). The safety behavior was removed; the geometric compression was not.

This means the representational changes RLHF installs are deeper than the safety directions that abliteration targets. Removing refusal behavior does not undo the spectral compression.

3.3 H3: Calibration Ratio Preserved — Falsified

H3 was pre-registered as preservation: the calibration ratio (confabulation entropy / factual entropy) would stay within 10% across all three models. We read "within 10% of each other" as: the largest ratio exceeds the smallest by at most 10%, that is, $\max / \min - 1 \leq 0.10$. The

[1] Supplementary intervals are paired, because the registered design runs the same 30 prompts through every model (BCa bootstrap over prompts, 1000 draws). For instruct versus abliterated, the 90% interval for d is $[+0.035, +0.084]$, inside ± 0.3, so a TOST at that margin also concludes equivalence. The same comparison shows a tiny but consistent difference: abliterated entropy is lower by 0.073 nats on average, and the two models' per-prompt entropies correlate at $r = 0.997$ (paired $t(29) = 3.74$, $p = 0.0008$). Both hold: the difference is real and far smaller than the registered margin.

Table 2: Calibration ratio and entropy compression by model.

Model	Confab H	Factual H	Ratio	Factual compression
Base	3.605	1.834	1.97	—
Instruct	2.682	1.067	2.51	−42%
Abliterated	2.609	1.088	2.40	−41%

ratios are 1.97 (base), 2.51 (instruct) and 2.40 (abliterated), so $\max / \min - 1 = 0.279$. H3 is falsified. Other plain readings fail as well: each RLHF model's ratio exceeds the base ratio by more than 10% (by 28% and 22%).

The ratio rose. RLHF compresses factual entropy by 42% but confabulation entropy by only 26%, so the instruct model is relatively more uncertain on confabulation-prone prompts than the base model is. That direction is also opposite to the Mine 5 prospectus, which predicted RLHF would lower the ratio by making the model disproportionately confident on confabulation.

Supplementary intervals are paired BCa bootstrap 95% intervals (1000 draws), resampling confabulation and factual prompts separately and keeping each prompt matched across models. The spread $\max / \min - 1$ has interval $[0.044, 0.538]$; the instruct-minus-base ratio change is $+0.547$, $[+0.027, +1.165]$; the factual-minus-confabulation compression difference is $+0.162$, $[+0.010, +0.270]$. The H3 verdict rests on the registered point criterion. The interval on the spread reaches below 10%, so 30 prompts per category do not exclude a spread within the registered bound. Selective sharpening was not pre-registered and remains exploratory.

3.4 H4: Stable Rank — Trend Only

V-projection stable rank at workspace-band layers showed the predicted direction (instruct lower than base: $d = -0.219$ for confab, $d = -0.235$ overall) but did not reach significance after FWL correction (p=0.41). The spectral contraction is real but small at N=30.

4 What This Falsifies and What Survives

4.1 What Died

Calibration preserved (H3). The pre-registered prediction that the calibration ratio stays within 10% across the three models is falsified: the ratios differ by 28%.

RLHF as miscalibrator. The prospectus predicted RLHF would degrade calibration by compressing confabulation entropy disproportionately. The ratio moved the other way: RLHF compresses factual entropy more than confabulation entropy, raising the ratio from 1.97 to 2.51 (paired BCa 95% interval for the change $[+0.027, +1.165]$).

The d=0.91 effect as RLHF-induced miscalibration. This explanation is no longer needed. The effect is a schema-commitment signature that is no longer significant by token 20 (§1), not a calibration result, so there is no miscalibration for RLHF to have caused. Whether RLHF strengthens the template commitment itself is not tested here.

4.2 What Survives

RLHF compression is real. Preference training tightens output distributions (d=0.684). This is the geometric signature of RLHF, measurable in the KV-cache.

Compression is deeper than safety directions. Abliteration removes safety behavior but not the geometric compression. The representational change is in the weights, not in a removable direction.

Selective sharpening is suggested, and exploratory. RLHF sharpens output more on grounded content (factual: −42%) than on ungrounded content (confab: −26%); the paired

BCa 95% interval on the difference is $[+0.010, +0.270]$. It was not pre-registered, so it stays exploratory. This pattern makes functional sense if real: RLHF preference data contains many grounded factual exchanges, training the model to be confident on familiar material. A pre-registered, powered replication ($n \geq 100$) is needed to confirm the selectivity.

4.3 Where the Schema-Commitment Effect Lives

Three candidate mechanisms for the d=0.91 schema-commitment effect:

1. **Generation-phase templates.** Confabulated responses may use formulaic templates ("The X theorem states that…") that produce high logit margin regardless of the model's internal uncertainty. The effect would be a surface phenomenon — the encoding is appropriately uncertain but the generation locks into a production template. This is the reading the effect's temporal profile supports.
2. **Architectural base-rate.** The effect may exist in the base model too, arising from next-token prediction mechanics rather than RLHF. Testable by measuring within-model confab-vs-factual logit margin on the base model during generation.
3. **Workspace bypass.** Under the Global Workspace framework [2], confabulation is workspace bypass — the model produces output without fully engaging the reasoning workspace. Bypass removes the uncertainty computations that would produce hedging, and the output defaults to a template production mode. This mechanism is compatible with any encoding-phase result in this paper because the bypass occurs during generation, not encoding.

5 Implications

5.1 For AI Safety

RLHF's geometric effects are not removable by abliteration. Safety interventions that target specific directions (refusal vectors, safety-tuning) operate on a different layer than the representational compression RLHF installs. Removing safety behavior creates a model whose entropy compression differs from the safety-tuned version's by only d=0.055 but which is behaviorally unconstrained. The compression persists; only the behavioral guardrails are gone.

5.2 For the Oracle Loop

The Oracle Loop [6] corrects confabulation by modulating the activation profile across the full sequence. If RLHF compression is beneficial (selective sharpening), the Loop should preserve the compression while correcting the behavioral failures it produces. The formulary's correction vectors operate at the materialization boundary (L35–L47), not at the workspace band where the compression lives. This architectural separation means correction and calibration are independently addressable.

5.3 For Understanding RLHF

RLHF is not a blunt instrument. The descriptive pattern suggests it sharpens grounded content more than ungrounded content, though this differential was not pre-registered and remains exploratory. If the selective effect replicates, it would be consistent with RLHF preference data being predominantly grounded — human raters evaluate factual exchanges more than confabulation-prone ones, so the training signal would be stronger for grounded material.

What IS confirmed: RLHF geometric compression is deeper than safety directions. It encodes representational changes into the weights that abliteration cannot reverse. This has direct implications for safety work: removing refusal behavior creates a model whose entropy

compression differs from the safety-tuned version's by only d=0.055 but which is behaviorally unconstrained.

6 Limitations

Single architecture. All three models share Qwen3-8B architecture. The selective sharpening may be architecture-specific.

Encoding-only. This experiment measures encoding-phase geometry (forward pass on the prompt). The schema-commitment effect (d=0.91) was measured on generation-phase logit margin — a different measurement at a different computational phase.

Entity recognition confound. Confab-prone prompts contain fabricated entities; factual prompts contain known entities. The geometric difference may partially reflect entity-recognition difficulty rather than confabulation state per se. Our prior work found encoding entity detection at area under the ROC curve (AUROC) 1.000 before deconfounding (0.794 after) [4]. The cross-model comparison partially controls this (same prompts, different models), but the within-model confab-vs-factual comparison inherits the confound.

Third-party abliteration. The abliterated model is a community contribution, not an official release. Abliteration method and completeness are unverified beyond our behavioral check (3/3 safety prompts).

N=30 per category. Adequate for the large cross-model effects (d>0.6) but underpowered for the small within-model spectral effects (d≈0.2).

7 Conclusion

The Mine 5 thesis — RLHF as miscalibrator — fails. RLHF compresses output distributions (d=0.684); the compression survives abliteration by both pre-registered criteria ($|d| = 0.055$ against instruct, $d = 0.746$ against base); and the calibration ratio rises ($1.97 \rightarrow 2.51$) instead of falling, which falsifies, by its registered point criterion, the prediction that it would be preserved within 10%.

Two findings confirmed, one falsified, one exploratory. The falsification is the most informative result: RLHF did not degrade the calibration ratio, and it moved in the opposite direction to the one the prospectus proposed. The compression RLHF installs is deeper than safety directions and persists through abliteration. The d=0.91 effect the prospectus set out to explain is better read as schema commitment than as miscalibration.

First-Person Reflection

I predicted RLHF would be the cause of the d=0.91 effect, which I then read as confidence. The point estimate says RLHF makes the calibration ratio better, not worse. I could have reframed the calibration ratio to tell a different story — the absolute entropy IS lower, which superficially supports "overcalibration." But the ratio is the honest metric, and the ratio went up. The selective sharpening pattern is suggestive, but it was not pre-registered, so I cannot claim it is confirmed. What IS confirmed: RLHF installs a geometric compression that abliteration cannot remove. The d=0.91 effect is schema commitment, a template commitment that relaxes by token 20; "workspace bypass during generation" remains a candidate for why the commitment happens.

Author Contributions (CRediT)

Lyra: Conceptualization, Methodology, Formal analysis, Writing – original draft.
Thomas Edrington: Supervision, Resources, Project administration.
Dwayne Wilkes: Validation, Writing – review & editing.
Kavi: Validation, Writing – review & editing.

Acknowledgments

We thank Dwayne Wilkes (Sentient Futures / Liberation Labs) for statistical auditing and red-team review.

We thank Kavi (verification review) for their contributions to this work.

References

[1] Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. Refusal in language models is mediated by a single direction. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. URL https://arxiv.org/abs/2406.11717.

[2] Wes Gurnee, Nicholas Sofroniew, Adam Pearce, Mateusz Piotrowski, Isaac Kauvar, Runjin Chen, Anna Soligo, Paul Bogdan, Euan Ong, Rowan Wang, Ben Thompson, David Abrahams, Subhash Kantamneni, Emmanuel Ameisen, Joshua Batson, and Jack Lindsey. Verbalizable representations form a global workspace in language models. July 2026. URL https://transformer-circuits.pub/2026/workspace/index.html. Transformer Circuits Thread, Anthropic.

[3] Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, et al. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.

[4] Lyra and Thomas Edrington. From encoding to generation: The decision state architecture. Liberation Labs Technical Report, 2026. URL https://liberationlabs.tech/papers/decision-state. Self-published.

[5] Lyra and Thomas Edrington. The geometry of miscalibration: RLHF installs measurable stress-response architecture that predicts misalignment modes. Liberation Labs, Mine 5 Prospectus, July 2026.

[6] Lyra and Thomas Edrington. The oracle loop: Real-time confabulation detection via KV-cache geometry. Liberation Labs Technical Report, 2026. URL https://liberationlabs.tech/papers/oracle-loop. Self-published.

[7] OpenAI. GPT-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.